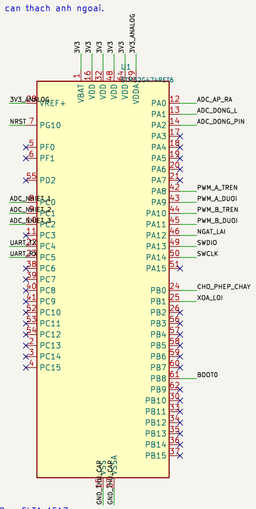
BO DIEU KHIEN TOI THIEU STM32G474RET6 / 3.3 V

Ban thiet ke nguyen ly de review va thu nguon thap. Chua co PCB, firmware hoan chinh hoac xac nhan tren phan cung.

1. MCU: dung HSI + PLL khi khoi dong; chua can thach anh ngoai.



PA8..PA11 = AF13 / HRTIM A1,A2,B1,B2. PA12 = FLT1 AF13.

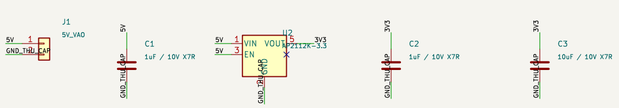
PG10/NRST (chan 7): giu che do RESET; khong doi thanh GPIO.

PB8/BOOT0 (61) keo xuong; kiem tra option byte boot Flash.

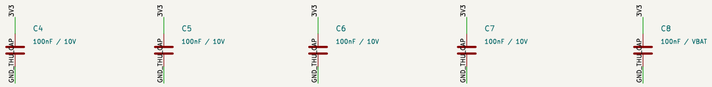
VREFBUF tat / high impedance khi VREF+ noi 3V3_ANALOG.

Moi ten net giong nhau duoc noi dien voi nhau; GND_THU_CAP khong phai DC-400V.

2. NGUON 5 V -> 3.3 V (chi nguon 5 V da on ap)

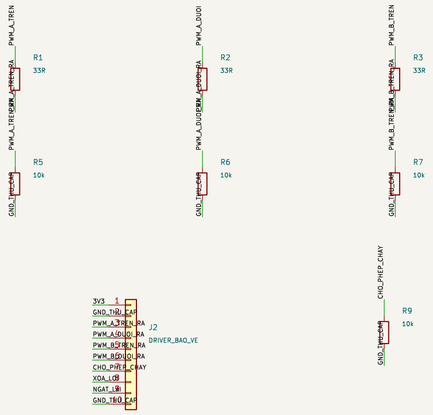


C1/C2 sat U2. Gioi han tai theo nhiet LDO: (5-3.3)*I: khong coi 600 mA la dong dung lien tuc o mot PCB.



C4..C7: moi tu sat mot VDD 16 / 32 / 48 / 64.

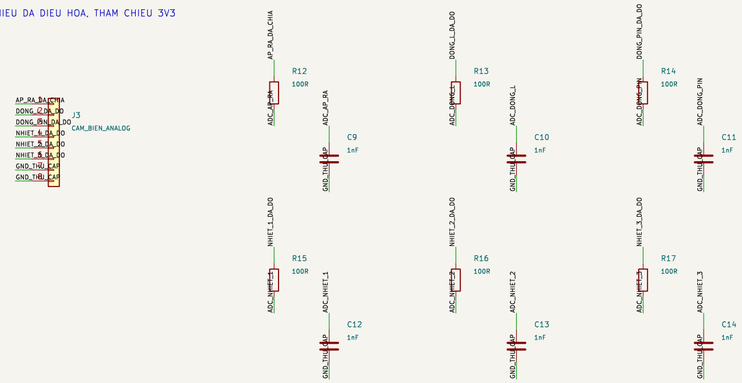
3. DAU NOI DRIVER / BAO VE - TIN HIEU 3.3 V



CHO_PHEP_CHAY=0 khi reset. XOA_LOI: chan du phong, chua co latch ngoai.

NGAT_LAI=1 => DIS/FLT1 (rut day => 1); khong dong nghia loi da duoc chot.

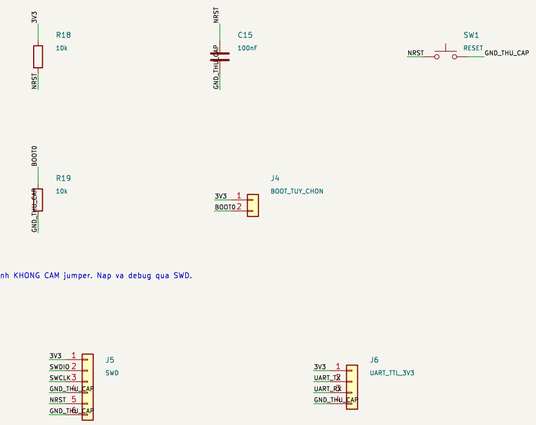
4. ADC: TIN HIEU DA DIEU HOA, THAM CHIEU 3V3



J3 chi nhan tin hieu da dieu hoa: RC 100R/1nF tai MCU bo sung cho RC tren bo do.

Kiem tra thoi gian lay mau ADC voi tro khang nguon; tat PWM khi dien ap/dong/nhiet vuot nguong.

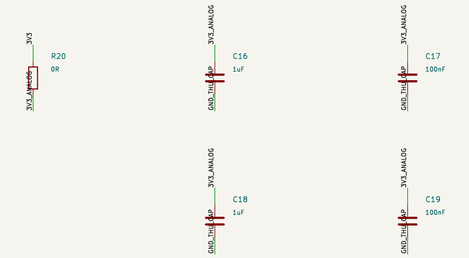
5. RESET / BOOT / SWD



J4 mac dinh KHONG CAM jumper. Nap va debug qua SWD.

J5.1 = VTREF cho ST-LINK; khong cap nguon tu 2 nguon cung luc.

6. NGUON ANALOG / THAM CHIEU



C16/C17 sat VDDA29-VSSA27; C18/C19 sat VREF+28-VSSA27.

Mat phang GND lien tuc; tach dong lon bang bo tri, khong cat mat phang duoi ADC.



PWR_FLAG: danh dau nguon ngoai / net sau R20 cho ERC; khong phai linh kien can mua.

LAYOUT VA THU LAN DAU

1. U1 LQFP64 chan 0.5 mm: han tren PCB; khong dung day Dupont cho duong cong suat/PWM driver.

2. Cap 5 V gioi han dong khi CHUA noi driver/cong suat. Do 3V3, VDDA va NRST truoc khi nap.

3. Test SWD, ADC va bon PWM bang oscilloscope. Kiem tra dead-time, khoa cung nhanh va FAULT truoc khi noi driver.

4. Firmware phai giu CHO_PHEP_CHAY=0 luc boot/reset/loi; chi cho chay sau khi nguon va cam bien hop le.

5. Nguyen ly nay khong bao gom mach bao ve cam bien, latch loi, nguon driver, duong cong suat va PCB400V.

6. Da doi chieu pin bang KiCad/ST open pin data. Chua lap rap hay do kiem thuc te; can review PCB truoc che tao.

Prototype controller - no high voltage on this board

Sheet: /
File: BO_DIEU_KHIEN_RET6.kicad_sch

Title: BO DIEU KHIEN RET6 - minimum 3.3V controller

Size: A2	Date: 2026-10-15	**Rev: 0.1 REVIEW**
KiCad E.D.A. 9.0.2		Id: 1/1